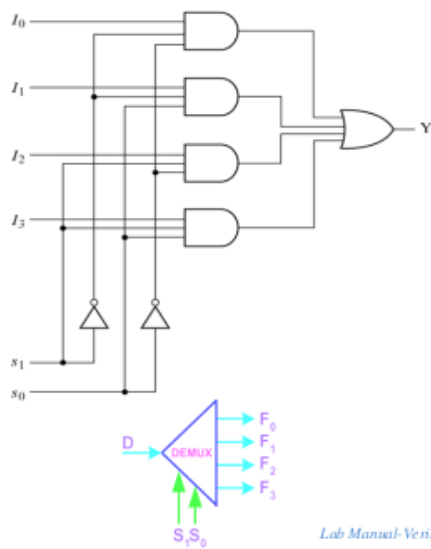


Circuit Simulation Lab:8

Demultiplexers

The purpose of this experiment is to introduce the design of simple combinational circuits, in this case Demultiplexer. Demultiplexer or Demux is a combinational circuit that distributes the single input data to a specific output line. The control inputs or selection lines are used to select a specific output line from the possible output lines.



1:4 Demultiplexer

s1	s0	F0	F1	F2	F3
0	0	0	0	0	0
0	1	0	0	0	0
1	0	0	0	0	0
1	1	0	0	0	0

Truth Table

Dataflow Modeling

Code:

```
module demux1_dataflow(
input d,
input s1,s0,
output y0,y1,y2,y3
);
```

```
assign y0 = d & ~s1 & ~s0;
assign y1 = d & ~s1 & s0;
assign y2 = d & s1 & ~s0;
assign y3 = d & s1 & s0;
```

```
Endmodule
```

```

module tb_demux1_dataflow;

reg d,s1,s0;
wire y0,y1,y2,y3;

demux1_dataflow dut(
.d(d),
.s1(s1),
.s0(s0),
.y0(y0),
.y1(y1),
.y2(y2),
.y3(y3)
);

initial begin

$dumpfile("demux1_dataflow.vcd");
$dumpvars(0,tb_demux1_dataflow);

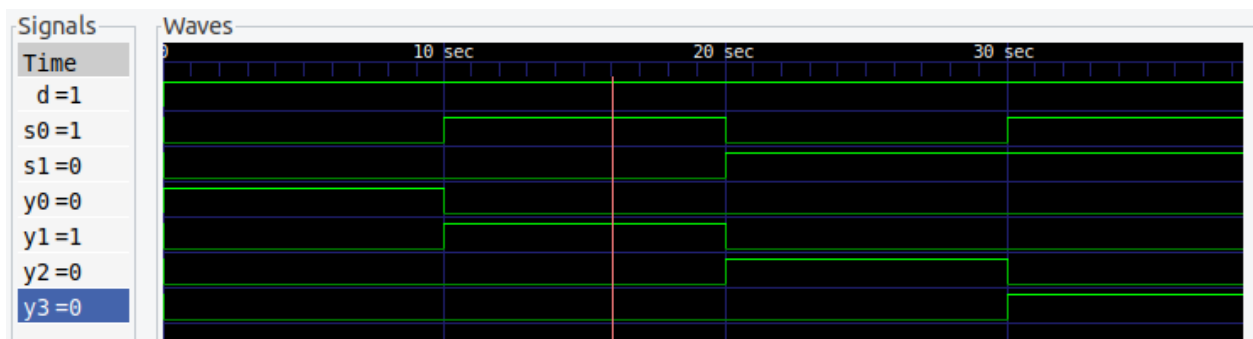
d=1;s1=0;s0=0; #10;
d=1;s1=0;s0=1; #10;
d=1;s1=1;s0=0; #10;
d=1;s1=1;s0=1; #10;

end

initial
$monitor("t=%0t d=%b s1=%b s0=%b y0=%b y1=%b y2=%b y3=%b",
$time,d,s1,s0,y0,y1,y2,y3);

endmodule

```



```

vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_8$ vvp
VCD info: dumpfile demux1_dataflow.vcd opened for output.
t=0 d=1 s1=0 s0=0 y0=1 y1=0 y2=0 y3=0
t=10 d=1 s1=0 s0=1 y0=0 y1=1 y2=0 y3=0
t=20 d=1 s1=1 s0=0 y0=0 y1=0 y2=1 y3=0
t=30 d=1 s1=1 s0=1 y0=0 y1=0 y2=0 y3=1

```

Behavioral Modeling

Code:

```

module demux1_behavioral(
input d,
input s1,s0,
output reg y0,y1,y2,y3
);

```

```

always @(*) begin

```

```

y0=0;
y1=0;
y2=0;
y3=0;

```

```

case({s1,s0})

```

```

2'b00 : y0=d;
2'b01 : y1=d;
2'b10 : y2=d;
2'b11 : y3=d;

```

```

endcase

```

```

end

```

```

Endmodule

```

```

module tb_demux1_behavioral;

```

```

reg d,s1,s0;
wire y0,y1,y2,y3;

```

```

demux1_behavioral dut(
.d(d),
.s1(s1),
.s0(s0),

```

```
.y0(y0),
.y1(y1),
.y2(y2),
.y3(y3)
);
```

```
initial begin
```

```
$dumpfile("demux1_behavioral.vcd");
$dumpvars(0,tb_demux1_behavioral);
```

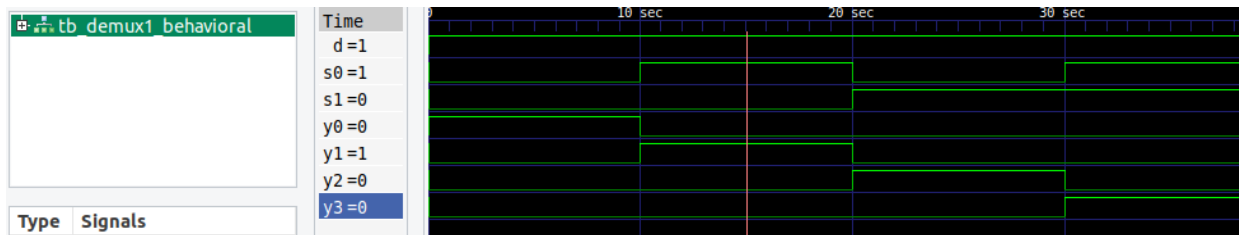
```
d=1;s1=0;s0=0; #10;
d=1;s1=0;s0=1; #10;
d=1;s1=1;s0=0; #10;
d=1;s1=1;s0=1; #10;
```

```
end
```

```
initial
```

```
$monitor("t=%0t d=%b s1=%b s0=%b y0=%b y1=%b y2=%b y3=%b",
$time,d,s1,s0,y0,y1,y2,y3);
```

```
endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_A
VCD info: dumpfile demux1_behavioral.vcd c
t=0 d=1 s1=0 s0=0 y0=1 y1=0 y2=0 y3=0
t=10 d=1 s1=0 s0=1 y0=0 y1=1 y2=0 y3=0
t=20 d=1 s1=1 s0=0 y0=0 y1=0 y2=1 y3=0
t=30 d=1 s1=1 s0=1 y0=0 y1=0 y2=0 y3=1
```

Structural Modeling

Code:

```
module demux1_structural(
input d,
input s1,s0,
output y0,y1,y2,y3
);
```

```
wire ns1,ns0;
```

```
not g1(ns1,s1);  
not g2(ns0,s0);
```

```
and g3(y0,d,ns1,ns0);  
and g4(y1,d,ns1,s0);  
and g5(y2,d,s1,ns0);  
and g6(y3,d,s1,s0);
```

```
Endmodule
```

```
module demux1_structural(  
input d,  
input s1,s0,  
output y0,y1,y2,y3  
);
```

```
wire ns1,ns0;
```

```
not g1(ns1,s1);  
not g2(ns0,s0);
```

```
and g3(y0,d,ns1,ns0);  
and g4(y1,d,ns1,s0);  
and g5(y2,d,s1,ns0);  
and g6(y3,d,s1,s0);
```

```
endmodule
```

